Lecture III - Packages and Data Management

Applied Optimization with Julia

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Quick Recap from last Week

Goals for Today

After this lecture, you will:

- Have refreshed the basics from last week's tutorials
- Know what packages are and why we use them
- Have seen your first DataFrame
- Be ready to start this week's tutorials

Variables and Data Types

- Variables are used to store values
- Assign a value to a variable using the `=` operator
- You can use different data types for variables
- You can change the value of a variable

. . .

> Tip
>
> You can use the `typeof` function to check the type of a variable.

Vectors and Matrices

- Vectors and matrices are used to store multiple values
- You can create them using the `[` and `]` operators
- Access their elements using square brackets

. . .

> Tip
>
> You can use the `push!` function to add elements to a vector or the `pop!` function to remove elements from a vector.

Comparisons and Logic

- Comparisons are used to compare values

- `==` checks if two values are equal
- `!=` checks if two values are not equal
- `<` and `>` check if one value is smaller or greater than the other
- `<=` and `>=` also allow the two values to be equal
- `&&` is true if both conditions are true
- `||` is true if at least one of two conditions is true

Loops

- Loops are used to repeat code
- `for` loop repeats code for a fixed number of times
- `while` loop repeats code as long as a condition is true

. . .

```
for product in ["apple", "banana", "cherry"]
    println("We sell: $product")
end
```

```
We sell: apple
We sell: banana
We sell: cherry
```

Conditionals

- Conditionals run code only under certain conditions
- `if` executes code if a condition is true
- `elseif` checks a further condition if the previous one was false
- `else` executes code if none of the conditions were true

. . .

> **i Note**
>
> Conditionals are not loops: they decide **whether** code runs, while loops decide **how often** it runs.

Dictionaries

- Dictionaries store key-value pairs, like a lookup table
- Access a value by its key using square brackets
- `keys` and `values` return all keys and values

. . .

```
student_ids = Dict("Elio" => 1001, "Bob" => 1002)
println("Elio's ID: ", student_ids["Elio"])
```

```
Elio's ID: 1001
```

. . .

Tip

Later in the course, we will use dictionaries to store and look up the data of our optimization models.

Solutions from last Week

- The solutions to the tutorials are made available at the end of each week
- You can access them in the project folder on GitHub
- Click on the GitHub icon (the Octocat) at the bottom right of the page

. . .

Tip

You can ask questions anytime in class or via email!

Packages and Data Management

Why Packages?

Question: **Do we have to write all code ourselves?**

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- Luckily not! We can use packages written by others
- Packages are collections of reusable code, e.g. `DataFrames.jl` for tables or `Plots.jl` for charts
- Julia's package manager `Pkg` installs them for us:

. . .

```
using Pkg
Pkg.add("DataFrames")
```

Data Management with DataFrames

- `DataFrames.jl` stores tabular data, like a spreadsheet in code

```
using DataFrames
sales = DataFrame(
    product = ["apple", "banana", "cherry"],
    price = [0.50, 0.30, 2.00],
    sold = [120, 90, 30]
)
```

```
3×3 DataFrame
 Row │ product  price    sold
     │ String   Float64  Int64
─────┼─────────────────────────
   1 │ apple        0.5    120
   2 │ banana       0.3     90
   3 │ cherry       2.0     30
```

...

- This week's tutorials show how to read, write, and plot such data

Five Tutorials for this Week

Topics of the Tutorials

- **Functions**: Learn how to define and use functions
- **Packages**: Learn how to install and use packages
- **DataFrames**: Learn how to work with tabular data in Julia
- **IO**: Learn how to read and write data in Julia
- **Plots**: Learn how to create plots in Julia

Get Started with the Tutorials

- Download this week's tutorials and start with the first one
- Remember, you can ask questions anytime!

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> i And that's it for this lecture!
>
> The remaining time we will already start working on this week's tutorials.

Literature

Literature

- Lauwens, B., & Downey, A. B. (2019). Think Julia: How to think like a computer scientist (First edition). O'Reilly®. Link to the free book website.
- Julia Documentation

For more interesting literature to learn more about Julia, take a look at the literature list of this course.

Bibliography